

BRNO-TURANY, Czech Republic

WMO: 117230

Lat: 49.1531N

Lon: 16.6889E

Elev: 246

StdP: 98.41

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.4	-15.6	1.0	-11.0	-12.9	1.3	-8.3	11.4	2.9	10.3	0.1	3.4	70	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	31.3	20.4	29.5	19.8	27.7	19.0	21.5	28.9	20.7	27.7	19.8	26.2	4.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.3	24.7	18.2	13.5	23.6	17.4	12.8	22.9	63.7	29.1	60.6	27.8	57.5	26.3	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.7	-14.2	34.0	3.4	1.8	-16.7	35.3	-18.7	36.4	-20.6	37.4	-23.1	38.7		
(5)			-14.7	23.2	3.3	1.0	-17.1	23.9	-19.0	24.5	-20.9	25.1	-23.3	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-1.2	0.7	4.9	10.4	14.9	18.5	20.3	20.2	15.2	9.9	4.8	-0.1	(6)
(7)		DBStd	8.58	4.53	4.50	4.17	4.18	3.56	3.59	3.30	3.40	3.40	3.80	4.13	4.19	(7)
(8)		HDD10.0	1346	346	261	164	44	4	0	0	0	2	50	161	312	(8)
(9)		HDD18.3	3291	604	494	418	238	117	42	18	19	104	262	405	570	(9)
(10)		CDD10.0	1319	0	0	5	58	156	254	321	317	158	47	6	0	(10)
(11)		CDD18.3	224	0	0	0	1	9	46	80	77	10	0	0	0	(11)
(12)		CDH23.3	2011	0	0	0	12	92	400	737	691	77	2	0	0	(12)
(13)		CDH26.7	599	0	0	0	1	11	106	243	227	11	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.4	3.6	4.0	4.0	3.7	3.3	3.3	3.1	3.4	3.3	3.4	3.2	(14)
(15)	Precipitation	PrecAvg	502	24	22	28	30	65	70	69	58	44	33	36	28	(15)
(16)		PrecMax	700	79	71	73	82	402	166	225	159	121	110	84	62	(16)
(17)		PrecMin	360	2	1	3	3	6	18	11	9	7	4	12	3	(17)
(18)		PrecStd	85	13	15	17	19	53	29	39	36	28	22	19	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	13.0	18.8	24.9	28.1	32.1	33.8	34.2	28.2	22.1	16.8	10.4	(19)
(20)			MCWB	8.3	8.5	10.9	15.6	18.2	21.2	20.9	21.0	18.6	15.8	12.7	8.2	(20)
(21)		2%	DB	8.2	10.5	16.2	22.2	25.8	29.6	31.3	31.3	25.5	19.4	14.0	8.1	(21)
(22)			MCWB	6.1	7.5	10.0	13.8	17.4	20.6	20.3	20.1	17.6	14.7	11.2	6.3	(22)
(23)		5%	DB	6.0	8.5	13.7	19.9	23.9	27.5	29.5	29.1	23.4	17.6	12.0	6.5	(23)
(24)			MCWB	4.3	6.0	8.4	12.6	16.5	19.3	19.6	19.6	16.7	13.7	10.0	5.1	(24)
(25)		10%	DB	4.5	6.6	11.5	17.9	21.8	25.6	27.5	27.1	21.5	15.7	10.4	5.1	(25)
(26)			MCWB	3.2	4.7	7.6	11.6	15.3	18.3	18.9	18.7	15.8	12.8	8.7	3.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	9.6	11.9	16.3	19.7	22.8	22.7	22.4	19.8	16.6	13.3	8.4	(27)
(28)			MCDWB	10.6	12.4	16.9	23.6	26.2	29.8	30.8	29.9	26.0	20.5	16.1	10.1	(28)
(29)		2%	WB	6.3	7.8	10.5	14.6	18.1	21.3	21.4	21.2	18.5	15.4	11.6	6.7	(29)
(30)			MCDWB	8.0	10.1	14.9	21.0	24.3	27.7	29.0	28.9	23.8	18.5	13.6	7.9	(30)
(31)		5%	WB	4.6	6.3	9.2	13.3	17.1	20.1	20.5	20.4	17.5	14.3	10.2	5.3	(31)
(32)			MCDWB	5.8	8.2	12.9	18.7	22.6	26.3	27.5	27.7	22.2	17.1	11.8	6.4	(32)
(33)		10%	WB	3.3	4.8	8.0	12.1	16.0	19.0	19.6	19.5	16.4	13.1	8.9	4.1	(33)
(34)			MCDWB	4.3	6.5	11.0	17.0	21.1	24.5	25.8	25.9	20.7	15.3	10.2	4.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	6.2	8.3	10.1	10.0	10.4	10.7	10.8	9.2	7.6	5.3	4.5	(35)
(36)			MCDBR	6.4	8.8	12.8	13.1	13.1	13.1	14.0	13.5	12.2	10.1	7.3	6.0	(36)
(37)			MCWBR	4.9	6.3	7.4	6.8	6.5	5.9	5.3	5.1	5.8	6.1	5.1	4.7	(37)
(38)		5% WB	MCDBR	6.1	8.4	11.1	12.0	11.7	12.1	12.4	12.4	10.9	9.3	6.7	5.7	(38)
(39)			MCWBR	4.9	6.3	7.1	6.4	6.3	6.0	5.3	5.2	5.6	5.9	4.9	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.337	0.377	0.423	0.422	0.422	0.434	0.430	0.401	0.373	0.349	0.312	(40)	
(41)		taud	2.340	2.317	2.257	2.166	2.208	2.244	2.217	2.242	2.291	2.341	2.354	2.376	(41)	
(42)		Ebn at Noon	731	802	825	821	839	840	823	806	791	747	678	682	(42)	
(43)		Edh at Noon	69	90	112	137	136	133	135	126	108	87	69	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.88	1.68	2.82	4.32	5.12	5.54	5.30	4.75	3.37	1.96	0.97	0.68	(44)	
(45)		RadStd	0.14	0.22	0.45	0.48	0.60	0.47	0.55	0.44	0.46	0.28	0.10	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.78	N/A	N/A	+1.08	N/A	N/A	-142	-207	+117	+60
(47) Regional (72 neighbors)		+0.55	N/A	N/A	+0.80	+0.42	+0.36	-128	-168	+77	+33

Nomenclature: See separate page